

Improving Evidence Use and Semantic Verification in Multi-Agent Business Analytics

A Failure-Guided Controlled Study

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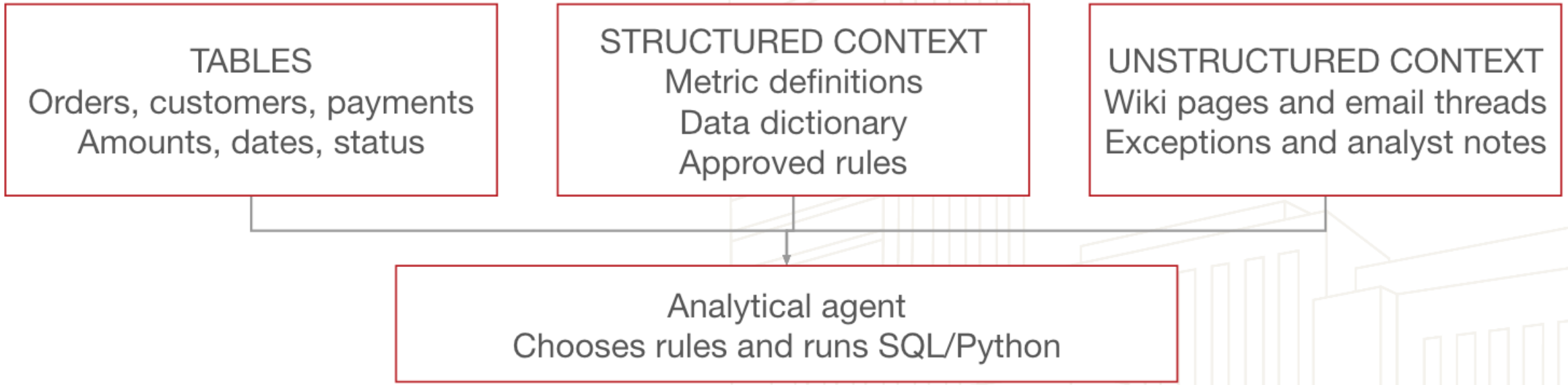
安泰经管学院
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Master of International Business
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Background

A Company Has Data and Business Context

A manager asks an analytical agent: “What were our May sales?”



The agent must connect the numbers to the applicable business meaning.

Background

The Same Tables Can Produce Different Sales Figures

Company rule: count completed orders in the month of completion.

Order	Amount	Order date	Completion date	Status
A	100	20 May	22 May	Completed
B	80	21 May	N/A	Cancelled
C	50	29 May	2 June	Completed

All orders placed in May
230

→

Completed orders only
150

→

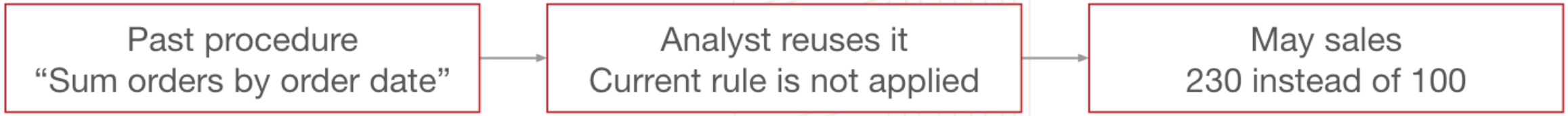
Completed during May
100

Illustrative company rule and data. Executing the code cannot establish which definition is intended.

Background

How the Wrong Interpretation Reaches the Final Answer

Problem 1: the analyst repeats a past calculation although today's rule differs.



Evidence-checked experience reuse: check the old procedure against current sources.

Problem 2: the reviewer checks execution and arithmetic, then approves 230.

The code runs. The sum is correct. The business definition was never adequately checked.

Source-grounded counterexample review: test the disputed interpretation.

Research Question

Reliability and Cost-effectiveness

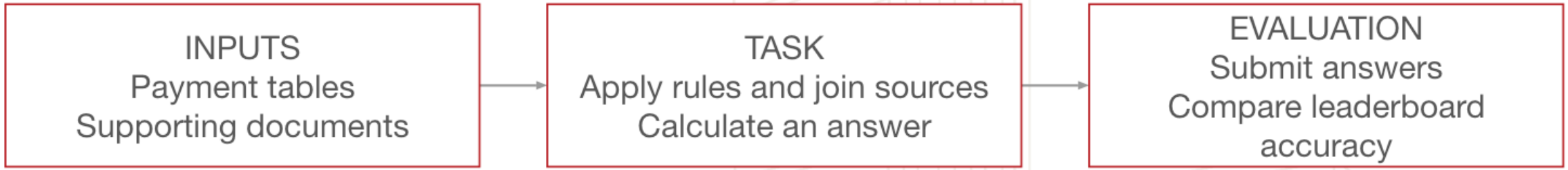
How do **evidence-checked experience reuse** and **source-grounded counterexample review** affect **reliability and cost-effectiveness** in a hierarchical multi-agent system for business analytics, **separately and together**?

Compare targeted failures and final answers.
Test added value against strong single-agent controls.

Literature Review

DABstep: A Shared Test for Data-analysis Agents

Data Agent Benchmark for Multi-step Reasoning
Public dataset and leaderboard by Adyen and Hugging Face



450 evaluation questions. Answers include numbers, short text and lists.
Example task type: identify applicable fee rules and calculate a merchant's fees.

DABstep paper; Hugging Face dataset; DABstep leaderboard

Literature Review

Existing Systems and Our Implementation Starting Point

System	What it provides	Role in this study
DataPilot	Versioned business assets and procedures	Knowledge reuse reference
DS-STAR	Planning, computation and iterative review	Review workflow reference
NVIDIA	Prepared tools with one agent at inference	Strong single-agent comparison
FinHelm	Memory, delegation and verification	Reproducible adaptation path

The first three are high-performing DABstep references in our 14 October audit.
FinHelm provides accessible infrastructure, not a claim to reproduce their scores.

DataPilot; DS-STAR v4; NVIDIA Data Explorer; FinHelm, pinned revision

Literature Review

MAST and Existing Approaches to the Two Problems

Cemri et al. (2025), "Why Do Multi-Agent LLM Systems Fail?"

MAST: Multi-Agent System Failure Taxonomy (14 failure modes).

System design	Inter-agent misalignment	Task verification

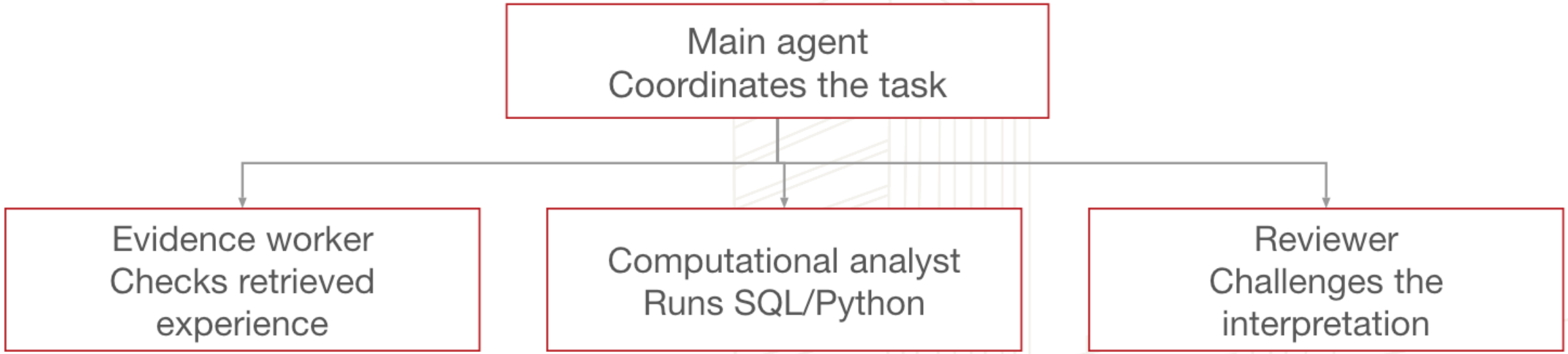
Problem and prior work	What our study will test
Experience use: DataPilot and ReasoningBank	Does past experience fit today's rules?
Review: SQLDriller, TS-SQL, QueryProof; DS-STAR	Does testing improve review decisions?

SQLDriller / TS-SQL: executable tests. QueryProof: business-rule gates.
MAST review failures: omitted checks (FM-3.2) and wrong judgements (FM-3.3).

Cemri et al., MAST v3; ReasoningBank v2; SQLDriller; TS-SQL; QueryProof v1

Research Plan & Method

A Fixed Hierarchy Adapted from FinHelm

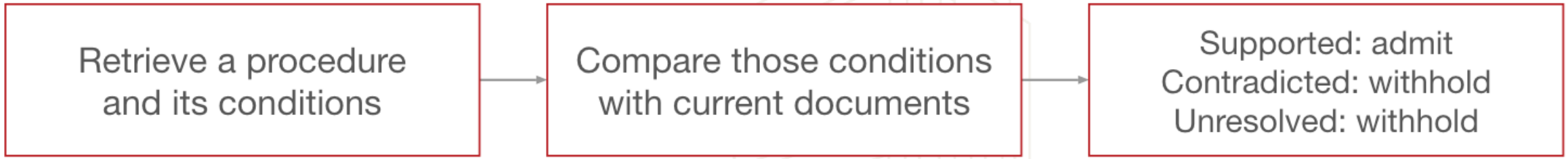


Separate working contexts and observable transfers.
The same roles remain in all four multi-agent conditions.

FinHelm, pinned revision; Scaling Agent Systems v3; Strong Single Agent Baseline

Research Plan & Method

Evidence-checked experience reuse



Past procedure: sum orders using order date.
Current rule: completed orders, using completion date.
Decision: withhold the incompatible procedure and analyse current sources.

Measure fewer inappropriate reuses without discarding useful experience.

DataPilot; ReasoningBank v2

Research Plan & Method

Source-grounded counterexample review

The reviewer reads the question and sources before seeing the candidate.

Small test case	By order date	By completion date
One order worth 50: placed 29 May, completed 2 June	May includes 50	May includes 0

The company rule requires completion date: this order contributes 0 to May.

Generate a valid distinguishing case, execute the candidate, then judge the result.
A wrong expected answer can still make the review fail.

SQLDriller; TS-SQL; QueryProof v1

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Research Plan & Method

Controlled Experiment and Evaluation

Condition	Experience check	Counterexample review
Baseline	Off	Off
Evidence-checked experience reuse	On	Off
Source-grounded counterexample review	Off	On
Both	On	On

Strong single-agent controls. Same model, sources, assets and aggregate caps.

DABstep measures final answers. A common candidate pool measures reviewers.
Analyse each mechanism separately and both together, with trace-based diagnosis.

Expected Results & Innovation Points

Research Contribution and Practical Use

Research contribution

Explain where business knowledge is lost or misapplied.

Test which mechanism changes the errors and final answers.

Provide a reproducible comparison and possible MAST refinements.

Practical use

Help analytics teams choose which checks to enable.

Show when preparing and reusing knowledge is worthwhile.

Clarify when a simpler single agent is sufficient.

Expected outputs: failure evidence, an implementation and quality–cost comparisons.

Expected Results & Innovation Points

When Is Additional Checking Worth Its Cost?

Operating cost

Model and tool spending per task
Latency, failed runs and retries
Cost per correct answer and coverage

Preparation and adaptation

Hours to prepare knowledge and memory
Effort to adapt to new tasks
Amortisation over future query volume
With explicit assumptions

Business decision

Compare quality at the same budget and cost at comparable quality.
Report when the added reliability justifies the extra work.

Challenges & Limitations

What May Limit the Benefits?

1. Analyst and reviewer may share the same wrong interpretation.
2. Generated tests can be invalid or have incorrect expected answers.
3. Useful experience may be rejected, or extra checks may cost too much.
4. Too few target failures can limit what the experiment can establish.
5. One hierarchy and benchmark do not represent every business setting.

Expected Research Progress

Prepared Work and Next Steps

Prepared: literature synthesis, problem map, research design and proposal.

Period	Planned work
Late October	Minimal harness and agreed model/budget
November	Implement, validate, pilot and freeze protocol
December	Main comparisons and midterm report
January	Failure, uncertainty and cost analysis
February–April	Thesis writing, revision and submission
May	Final defense

The outcome is evidence about which checks are worth using in business analytics.